Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Consider two random processes X_k and Y_k where we compute the value $Y_k = 0.8X_k + 0.2X_{k-1}$. If X_k is a white-noise process, what can we say about Y_k?

1 / 1 point

- ○ Y_k is also a white-noise process.
- ◉ Y_k cannot be a white-noise process.
- ○ Y_k may or may not be a white noise process.

Correct
Correct. We can compute

$R_Y(0) = 0.68.$

$R_Y(\pm 1) = 0.16.$

$R_Y(\tau) = 0$ for all other values of τ.

Since this is not equal to a constant multiplying $\delta(\tau)$, Y_k cannot be a white-noise process.

2. Consider two random processes X_k and Y_k where we compute the value $Y_k = X_k + 1$. If X_k is a white-noise process, what can we say about Y_k?

1 / 1 point

- ○ Y_k is also a white-noise process.
- ◉ Y_k cannot be a white-noise process.
- ○ Y_k may or may not be a white-noise process.

Correct
Correct. We can find that $\mathbb{E}[Y_k] = \mathbb{E}[X_k + 1] = 1$. All white-noise processes have a mean of zero, and so Y_k cannot be a white-noise process.

3. Which of the following statements are always true about Gaussian random processes. Select all that apply.

1 / 1 point

- [x] They can be reasonable models of noises that are themselves caused by the summation of many independent random causes.

Correct
Yes. By the central limit theorem (discussed in lesson 3.1.6), the sum of many independent random quantities converges to a Gaussian distribution. So, a Gaussian process can be a good model for these kind of noises.

- [x] The pdf of the random process at any point in time can be completely specified by the mean and covariance of the process at that point in time.

Correct
Yes. A Gaussian random variable is completely specified by its mean and covariance. Since every sample of a Gaussian random process is a Gaussian random variable, this assertion is true.

- [ ] Gaussian random processes are always white.
- [ ] Gaussian random processes are always stationary.